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Boundary-Aware Semantic Image Inpainting with Spectral Coherence Supervision

 **NeurIPS**

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Summary

The paper targets a well-known failure mode in semantic image inpainting: visible artifacts at the seam where synthesized content meets valid context. It proposes two localized supervision mechanisms—a gradient-weighted boundary loss and a localized spectral boundary coherence loss—together with mask-injected attention skip gates and gated convolutions in a U-Net-like architecture. A four-phase ablation on Places365 and a zero-shot test on CelebA-HQ suggest improved boundary fidelity (as measured by a proposed Boundary-MAE and a spectral score) with a slight trade-off in global pixel-level metrics.

Strengths

Technical novelty and innovation

- The idea of localizing spectral supervision to patches intersecting

the inpainting boundary is a thoughtful and targeted contribution distinct from global frequency losses such as FFL.

- The gradient-weighted boundary loss focuses learning capacity on high-frequency transitions near seams, aligning with the problem statement.
- The mask-injected attention skip gate provides a lightweight, plausible mechanism to modulate skip connections using mask information, mitigating leakage from corrupted encoder features.

Experimental rigor and validation

- The paper follows a clear four-phase ablation that isolates contributions from backbone depth, skip gating, convolution type, and loss terms.
- Zero-shot cross-domain evaluation on CelebA-HQ offers preliminary evidence of transferability of the learned boundary priors.

Clarity of presentation

- The problem is well-motivated with a clear hypothesis (global losses underweight boundary pixels).
- The training objective and architectural choices are explained with reasonable detail and intuition.

Significance of contributions

- Addressing boundary seams is practically important for inpainting applications, and the proposed losses are general and could be adopted in other architectures.
- Introducing a boundary-focused metric (Boundary-MAE) helps measure the intended effect more directly than standard global metrics.

Weaknesses

Technical limitations or concerns

- The spectral loss uses only FFT magnitudes on small patches, discarding phase information that can be critical for edge alignment; the implications and alternatives (e.g., complex or log-

magnitude comparisons) are not discussed.

- Randomly sampling only $K=8$ boundary-intersecting patches per image may provide sparse and potentially unstable supervision; no analysis of K or patch size sensitivity is provided.
- The gradient-weighted boundary loss risks overweighting noisy gradient regions; normalization and robustness measures are insufficiently detailed.

Experimental gaps or methodological issues

- No comparisons to strong, modern baselines (e.g., LaMa/Fourier convolutions, AOT-GAN, CoModGAN, EdgeConnect, MAT, Restore/Inpaint diffusion methods) are provided, even when some are referenced; conclusions remain within-architecture only.
- Reported improvements in Boundary-MAE are small and lack confidence intervals/significance tests; the spectral coherence metric is insufficiently defined to interpret the 0.983x numbers.
- Cross-domain results do not include boundary-focused metrics or perceptual metrics (LPIPS), weakening the transferability claim specifically for seams.
- Training on a nonstandard subset of Places365 for only 15 epochs may limit conclusions; no compute/training time or convergence curves are reported for the final setting.

Clarity or presentation issues

- Some loss definitions are under-specified: dilation radius r for boundary band, normalization of gradient map, exact spectral coherence score, and the multi-scale loss formula.
- The scaling factor α for the boundary loss is mentioned but not included in the final formula.
- Image resolution, augmentation details, and mask distribution for evaluation are not fully specified.

Missing related work or comparisons

- No direct empirical or analytical comparisons to frequency-domain inpainting methods (e.g., LaMa) or local boundary-aware priors (e.g., edge-aware/gradient-domain blending) are included.
- User/perceptual studies or LPIPS/FID are missing, which are common in modern inpainting evaluation.

Detailed Comments

Technical soundness evaluation

- The localized boundary objectives are well-motivated and technically plausible. However, the spectral loss's reliance on FFT magnitudes only and a small number of patches raises questions about sensitivity to phase misalignment and local shifts. A discussion contrasting magnitude-only vs. complex-valued losses, and a grid-search over K and patch size, would strengthen soundness claims.
- The mask-injected attention skip gate is sensible and consistent with mask-aware design trends, but the relation to existing gated/attention-based skip variants (e.g., learned gating of skip connections, SPADE-like modulation, or CBAM-style attention) deserves deeper contextualization.
- The architecture excludes adversarial training, which is common for sharper inpainting, yet the claims emphasize perceptual seam quality; an explicit rationale for omitting adversarial losses would be useful, as would demonstrating that the proposed losses can complement GAN-based setups.

Experimental evaluation assessment

- The four-phase ablation is systematic and helpful. Still, the final claims would benefit from:
 - Strong baseline comparisons (at minimum: Partial Convs, Gated Convs baselines with standard losses, LaMa/FFTConv, and a frequency-loss baseline such as FFL).
 - Perceptual metrics (LPIPS), human studies, and boundary-focused user preference tests, given the emphasis on seam realism.
 - Statistical reporting (mean $\pm$ std over multiple seeds, or bootstrapped CIs) for Boundary-MAE and spectral metrics.
 - Ablations over boundary band width r , number of spectral patches K , patch size, and weighting coefficients.
- The zero-shot evaluation should include the boundary-specific metrics to substantiate the transfer of seam improvements.

Comparison with related work (using the summaries provided)

- Relative to Partial Convs and Gated Convs, the main novelty lies in boundary-focused supervision and mask-aware gating of skips rather than the base convolution operator.
- Compared to Focal Frequency Loss and LaMa's Fourier convolutions, the localized spectral supervision is an interesting twist. However, without direct comparisons to FFL and LaMa, it is difficult to assess whether the localized spectral constraint yields superior boundary fidelity in practice.
- Prior work using edge maps/structure guidance (e.g., edge-aware inpainting) also targets transitions; referencing and/or comparing with such edge-conditioned approaches would contextualize the gradient-weighted boundary loss more fully.

Discussion of broader impact and significance

- If validated against strong baselines, localized seam supervision could be valuable in photo editing, AR/VR, and cultural heritage restoration where seams are highly distracting. Conversely, improved seam invisibility also increases the risk of undetectable image manipulation; a short ethics note on potential misuse and detection countermeasures would be appropriate.
- The proposed boundary metrics could stimulate more nuanced evaluation of inpainting quality beyond global pixel metrics.

Questions for Authors

1. How is the Spectral Coherence Score computed precisely (normalization, frequency weighting, per-patch aggregation, range/interpretation of reported $\sim 0.983x$ values)?
2. What dilation radius r defines the boundary band B , and how sensitive are results to r ? Please include an ablation over r .
3. Why restrict the spectral loss to magnitude-only FFTs? Did you evaluate complex-valued or log-magnitude variants, or phase-sensitive alternatives?
4. How sensitive are results to the number of spectral patches K and patch

size? Did you try deterministic coverage of the boundary band instead of random sampling?

5. The α scaling factor is mentioned in the boundary loss text but not in the formula. Is α used, and if so, how is it set/normalized?
6. What is the input resolution and exact preprocessing/augmentation pipeline? Are masks in the test set matched in coverage/shape distribution to training masks?
7. Could you compare L4 against FFL and LaMa under identical data/masks to situate your contribution relative to frequency-domain SOTA?
8. Did you try adding an adversarial discriminator and, if so, does the boundary/spectral supervision remain beneficial?
9. Can you report LPIPS and, if possible, a small-scale user study or boundary-focused preference test?
10. In the CelebA-HQ zero-shot, can you include Boundary-MAE and the spectral coherence score to support the claim that seam improvements transfer?

Overall Assessment

The paper addresses an important and under-explored aspect of semantic inpainting—seam artifacts at the inpainting boundary—with a clear hypothesis and targeted losses. The localized spectral coherence loss is a neat idea, and the mask-aware skip gating is a reasonable architectural refinement. The ablations within the proposed framework are systematic and suggest small but consistent improvements on boundary-focused metrics, with a trade-off in global PSNR/Hole-MAE. However, the work falls short of NeurIPS standards in experimental validation: there are no comparisons to strong, modern baselines, boundary-focused gains are numerically modest without statistical analysis, and several key definitions and hyperparameters are under-specified. Strengthening the empirical section with mainstream baselines (e.g., LaMa, FFL), perceptual metrics, boundary-focused cross-domain evaluation, and clearer metric definitions would substantially improve the paper. As it stands, I view the contribution as promising but preliminary; I recommend rejection in the current form, with a clear path to a stronger submission after addressing the above

issues.

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